

Fifth workshop on Metascience for Machine Learning

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UNIT
DELFT



AI Initiative

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Renee Koster.

What is Metascience for Machine Learning?

Holding a magnifying glass up to the ways of doing ML research

Metascience for machine learning focuses on the science in the field of machine learning. It's about topics that are typically not found in Machine Learning text books.

It's about asking how and why research in machine learning is done in certain ways, the methodology, the processes, the mindset, goals, aspirations, inspirations.

At its heart, metascience for ML seeks to critically evaluate, understand, and enhance the way we generate, assess, and communicate scientific knowledge in machine learning—guiding the field toward more rigorous, reflective, and meaningful progress.

Goals of Metascience for ML

1. Scoping:

What is metascience for machine learning formalised via written (joint) perspectives

2. Strategic Development:

What activities could be helpful to address the problem?

What are the low hanging fruit,

where are opportunities for additional seed funding/catalysers?

3. Who and What:

who is available to contribute what in the developed strategy?

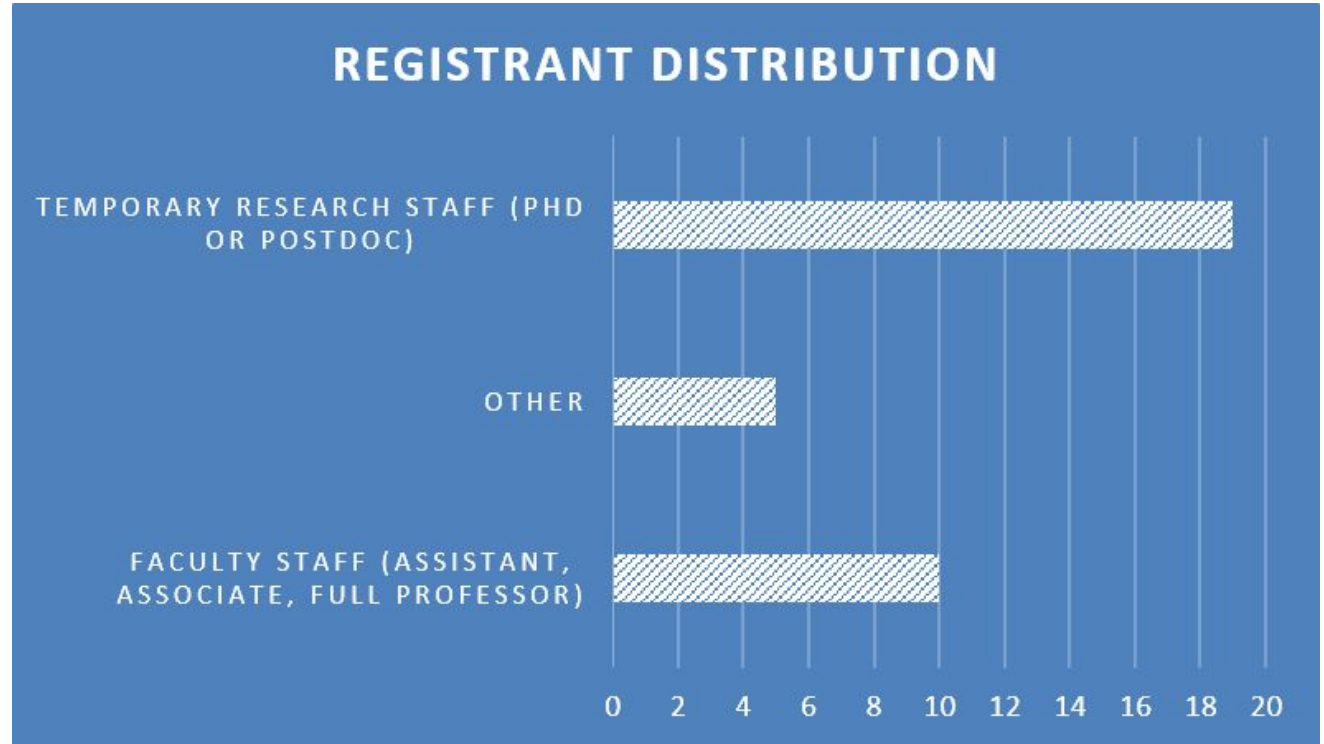
Making the “what” **desirable and manageable**

What has happened so far?

- 4 Workshops since June 2025
 - W1: Kick off, W2: metascience of benchmarking, W3: metascience in other disciplines, W4: Empirical Theory in ML
- Working groups: What is Theory in ML? Measuring the Problem
- Web page and social media
- 5 Podcasts : Proposition format: launching before the end of the year
 - **Live audience , roughly bi monthly Wednesday lunchtimes in VMB**
 - Next one: tomorrow at 11:30-13:00 Ritchie (5th floor). Feel free to dip in and dip out :)

Registration Stats today

- ~35 signups



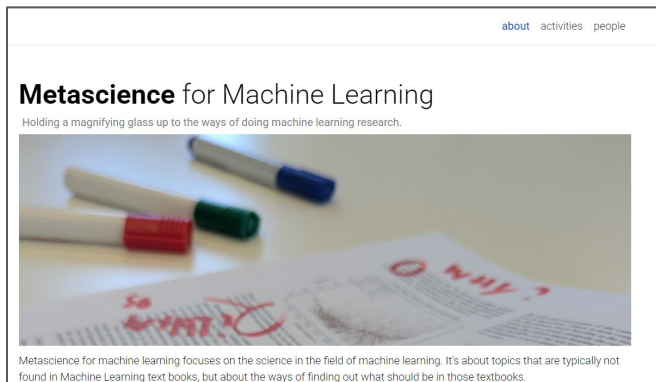
Community Introductions

- [Offline Intro Slides For Workshop Attendees - Google Slides](#) : Introduce yourself!
- [How do you identify as a ML researcher](#)? Put yourself in the How / Why space

Communication Channels

Web page

<https://metascienceforml.github.io/>



People page needs more people

June 2025	We organized our first workshop on metascience for machine learning on June 20, 2025! 🌟
March 2025	We got a starting fund from Ellis unit Delft 😊

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Mailing list:

metascience4machinelearning-ewi@lists.tudelft.nl



Schedule

• 10:00: Workshop Introduction (Hayley)	Chair: Jan
• 10:10: Abigail Nieves Delgado and Marilou Niedda (University of Utrecht), “Philosophy of Science as a Tool for Scientific Practice”. They will also lead a Group Activity on Fuzzy Cognitive Maps	Chair: Hayley
• 12:40: Lunch	
• 13:30: David Tax	Chair: Jan
• 14:15: Closing	Chair: Marco

Lunch head count!

House Rules

- **Be mindful of sub-disciplinary culture:**
 - what you consider basic knowledge may be entirely unknown to your conversation partner.
- **Be prepared to play devil's advocate**
- **Avoid drowning in details:**
 - Identify key problems at the abstract level
- **Be aware of our stakeholders:**
 - ELLIS, ML community,?
- **Be mindful that strategy is implemented by concrete action:**
 - For each abstracted problem, have
 - short term easy goals (2025)
 - longer term more ambitious goals 2026-...?